

Self-Critiques

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Project Scope and Research Direction

This project began as a dynamic equity portfolio optimization project and later expanded into prediction markets through Polymarket and Kalshi. That expansion made the project more original and intellectually interesting, but it also introduced new challenges in data construction, validation, and interpretation. In retrospect, the project is best understood not as a finished trading system, but as an exploratory study of how numerical optimization methods behave when applied to noisy, incomplete, and difficult-to-validate market data. At the beginning of the semester, our problem formulation was too broad. We started with the general ambition of building a model that could perform better than existing stock-return models. Over time, we narrowed this into a more concrete dynamic portfolio optimization problem with rolling estimates, long-only constraints, and turnover penalties. This narrowing was important because it gave the project a mathematical and computational structure. However, we did not define the evaluation target early enough. We moved between several possible goals, including return prediction, portfolio Sharpe ratio, cumulative return, benchmark outperformance, robustness across hyperparameters, and eventual extension to prediction markets. If we were restarting the project, we would define the primary evaluation metric and validation scheme before building most of the pipeline.

One strength of the project is that it became a real implementation rather than a purely conceptual exercise. We built return panels, estimated rolling moments, optimized weights using PyTorch, evaluated out-of-sample portfolio returns, and compared the results to an equal-weight benchmark formed on the same date-specific universe. The monthly equity model was the most complete version of this pipeline. It had a clear objective function, a dynamic universe, rolling-window estimation, and benchmark comparison. The project therefore succeeded in turning portfolio choice into a numerical optimization problem that could be coded, tested, criticized, and extended. At the same time, the equity portion relied heavily on historical sample means and covariances, which are known to be noisy in financial data. We partially addressed this problem through rolling windows, ridge regularization of covariance matrices, turnover penalties, and sensitivity analysis. Future work should compare sample covariance estimates against shrinkage covariance estimators, factor-model covariance structures, exponentially weighted estimates, and more heavily regularized expected-return estimates. There are two considerations about our hyperparameter search. On the positive side, we did not simply choose one parameter setting and report it. We explored rolling windows, learning rates, gradient steps, risk-aversion parameters, and turnover penalties. This helped us see that the optimizer was responsive to hyperparameter choice and that small modeling choices could substantially change the realized portfolio. This was one of the most important lessons of the project: the objective function is not the whole strategy. The realized strategy also depends on the regularization strength, estimation window, initialization, solver configuration, and rebalancing procedure. One qualification is that our hyperparameter exploration should not be interpreted as ordinary in-sample fitting. In the equity portion, the portfolio rule was evaluated through a rolling-window design: at each date, the optimizer estimated moments using only prior returns and then evaluated the resulting weights on subsequent returns. This structure already imposed chronological discipline and reduced the most direct form of lookahead bias. The main limitation was instead practical. Because the monthly equity sample contained relatively few portfolio for-

mation dates, a strict division into separate development, validation, and untouched test periods would have left each segment fairly small and potentially unstable. With a longer time series, especially in the daily setting, a stronger design would reserve an explicit validation period for choosing hyperparameters and a final untouched test period for reporting performance. Thus, the concern is not that the rolling backtest was invalid, but that future versions with more date-level observations could impose an even cleaner separation between model exploration, calibration, and final evaluation. The benchmark comparison was necessary, but incomplete. We were right to compare the optimized portfolio to an equal-weight portfolio on the exact same date-specific universe, because this was the cleanest internal benchmark for isolating the contribution of the optimization rule itself. We also compared the strongest monthly strategy against broad market benchmarks such as the S&P 500, Nasdaq 100, and Russell 2000, which helped place the results in a more familiar economic context. However, those index comparisons are not perfect substitutes for portfolio-rule baselines, because the benchmark indices differ in universe construction, weighting scheme, and possibly timing. Future research could therefore include additional internal baselines, such as market-cap weighting, minimum variance, risk parity, momentum, and simple volatility-scaled strategies, all constructed on the same date-specific universe. These comparisons would make it easier to determine whether the optimizer is adding value beyond standard portfolio heuristics rather than only outperforming equal weighting or broad passive indices. The daily-data extension was useful because it tested whether the same optimization logic could survive a higher-frequency setting. Moving from monthly to daily data exposed new difficulties: daily returns are noisier, transaction costs matter more, rebalancing frequency becomes more consequential, and covariance estimation becomes more delicate. We rescaled the rolling window, minimum-history requirement, and turnover penalty to make the daily model more coherent, but we did not fully redesign the methodology for daily trading. A future version should examine less frequent rebalancing on daily data, explicit transaction-cost models, volatility targeting, liquidity filters, and execution-aware turnover constraints.

Prediction-Market Pivot

The biggest conceptual improvement in the project was the pivot toward prediction markets. After realizing that the equity optimizer was close to standard portfolio-optimization projects, we expanded the framework to Polymarket and Kalshi. This was a strong direction because prediction markets still fit the same broad allocation question: how should capital be distributed across risky opportunities when prices, probabilities, uncertainty, and liquidity constraints all matter? Instead of asking whether a stock has positive expected return, the prediction-market version asks whether a contract appears mispriced relative to an estimated probability and an executable market price. This extension gave the project a more original second half. It forced us to reinterpret portfolio optimization outside the usual equity setting. In equities, returns are claims on securities whose prices evolve over time. In prediction markets, prices are closer to market-implied probabilities of discrete future events. That difference changes the meaning of the optimization output. A prediction-market optimizer is not only selecting assets; it is testing whether a probability model can identify mispriced contracts after accounting for liquidity, bid-ask spreads, concentration limits, and cash constraints. However, the prediction-market pivot also revealed one of the largest limitations of the project: data availability. A clean prediction-market backtest requires historical prices, timestamps, liquidity information, order-book depth, resolved outcomes, and realistic assumptions about execution. Our available Polymarket data was closer to a structured market snapshot and daily artifact than a fully resolved historical panel. This meant that we had to be careful about what the project could claim. We could build a coherent baseline pipeline, study pricing and liquidity features, construct chronology-safe diagnostics, and examine quote-to-quote behavior. But we could not honestly claim to have built a fully validated prediction-market trading strategy. The Polymarket data exploration was therefore one of the most important parts of the project. It forced us to distinguish between data that looked useful and data that could actually support out-of-sample economic claims. Fields such as volume, liquidity, bid-ask information, outcome prices, event meta-

data, and timing variables were useful for feature construction and descriptive analysis. But without a complete resolved historical panel, realized PnL remained difficult to interpret. This was a valuable lesson: in prediction markets, the challenge is not simply estimating probabilities, but matching each prediction to the correct market price, execution condition, liquidity state, and eventual resolution.

Our baseline prediction-market pipeline was a meaningful extension of the original optimizer, even though it remained preliminary. We adapted the allocation objective so that the reward term became estimated edge: the difference between our model-implied probability and the conservative executable contract price. We also added constraints that are natural for prediction markets, including cash holding, long-only contract positions, per-contract caps, per-event caps, and liquidity-based position limits. These constraints made the optimization problem more realistic than a simple ranking exercise. Still, the prediction-market model was intentionally simple. That simplicity was useful because the baseline was transparent and easier to debug. It allowed us to focus on whether the allocation framework could be transferred into a new market setting. But it was also a limitation because the probability model itself was not very sophisticated. Future work should build richer features from market text, event categories, price history, volume changes, order-book depth, news signals, and possibly external fundamentals. Natural language processing could be especially useful because prediction-market contracts are defined by text, and the semantic content of the question is often central to whether the market is mispriced.

Validation, Interpretation, and Limitations

Our evaluation of the prediction-market portion was intentionally cautious. We used walk-forward methods to avoid obvious lookahead bias, and we examined diagnostics such as ranking behavior, allocation behavior, liquidity exposure, turnover, and next-step quote-to-quote PnL. This was the right direction because a single backtest number can be misleading when the data is incomplete. The daily re-optimization pipeline was especially useful because

it forced the model to repeatedly update using only earlier observations and then evaluate subsequent quote movements. The most important limitation is that quote-to-quote PnL is not the same as resolution-based trading profit. Because the packaged artifact did not include final contract outcomes, realized PnL had to be measured using the next observed implied price. This is still informative: it tests whether the model’s estimated edge predicts near-term market repricing. But it does not fully answer whether the strategy would make money when contracts settle. Future research could build a resolved event-level dataset and evaluate profit using final binary outcomes, while also accounting for bid-ask spreads, slippage, trading fees, and market depth. Another limitation was timing. The prediction-market pivot made the project more interesting, but it happened late enough that the implementation could not become as mature as the equity pipeline. If we had pivoted earlier, we could have spent more time collecting historical Polymarket or Kalshi data, designing a cleaner event-resolution pipeline, and building a stronger feature set. Instead, the project ended with a more developed equity optimizer and a promising prediction-market extension. This imbalance is important to acknowledge. As such, the prediction-market section should be interpreted as a baseline research framework and future direction, not as a final trading model.

Our development process was iterative. We moved through cycles of observing results, identifying problems, changing code, rerunning experiments, and revising the report. This was especially true during the prediction-market phase, where parsing nested data, organizing time splits, debugging file paths, and interpreting liquidity constraints required repeated revision. The use of LLMs was helpful, but it also introduced risks. LLMs accelerated code scaffolding, debugging, explanation, slide writing, and self-critique drafting. They were especially useful during the Polymarket phase because they helped us impose structure on unfamiliar data and quickly generate starter code for exploration, feature construction, and diagnostics. However, LLM outputs often sounded more complete than the underlying evidence allowed. For example, an LLM could suggest a full prediction-market backtesting

pipeline, but we still had to decide whether our data actually supported that pipeline. The project improved when we treated LLMs as assistants for iteration rather than as authorities. The repository organization improved over time and we hope it will allow other interested parties to reproduce our results and continue the related research including the research directions enumerated above and below.

Future Research Directions

Several future research directions follow naturally from the limitations of the project. The first is to build a stronger equity optimizer using better estimates of expected returns and risk. Instead of relying primarily on rolling sample means and covariances, future work should test shrinkage covariance estimators, factor-model covariance matrices, Bayesian expected-return estimates, volatility scaling, and transaction-cost-aware objectives. This would make the equity results less dependent on raw historical moment estimates. The second direction is to strengthen the evaluation design. Dependent on more historical data availability, A future version should use a fully pre-specified walk-forward validation structure with separate development, validation, and test periods. Hyperparameters should be selected only on validation periods, and the final reported performance should come from untouched test periods. This would reduce the risk of backtest overfitting and make the results more credible. The third direction is to expand the benchmark set. In equities, the optimized portfolio should be compared not only to equal weighting and passive market benchmarks, but also to market-cap weighting, minimum variance, risk parity, momentum, and simple factor-based strategies. In prediction markets, the model should be compared to market-implied probabilities, liquidity-only rules, volume-only rules, category-level baselines, and simple no-trade or cash benchmarks. These comparisons would help identify whether the optimizer is adding value beyond simple heuristics. The fourth direction is to build a resolved prediction-market dataset. This is probably the most important next step for the prediction-market portion. A stronger dataset would link each contract to a full price history, order-book

information, liquidity measures, timestamps, event category, final resolution, and settlement date. With that structure, the project could evaluate realized settlement-based PnL rather than only next-step quote changes. And, finally the fifth direction is to make the prediction-market optimizer more execution-aware. Future work should model bid-ask spreads, market depth, partial fills, slippage, fees, and order timing. It should also distinguish between theoretical edge and executable edge. A contract may appear mispriced at the midpoint but become unattractive once the model accounts for the best ask, depth available at that price, and the cost of exiting the position.

Overall Assessment

Overall, the main contribution of the project is not that it produced a finished or reliably profitable trading strategy. That would be too strong a claim. The main contribution is that we built, tested, and critically evaluated a flexible numerical optimization framework for portfolio allocation across two different market settings. In the equity portion, we implemented a dynamic optimizer with rolling estimation, long-only constraints, turnover penalties, PyTorch-based gradient optimization, and meaningful benchmark comparisons. In the prediction-market portion, we adapted the same allocation logic to a setting where prices represent market-implied probabilities rather than conventional asset values. This extension forced us to confront a different set of modeling issues, including executable prices, liquidity limits, event-level concentration, incomplete resolutions, and chronology-safe evaluation. This makes the project a good fit for the goals of the course. The course emphasized not only writing down optimization objectives, but also implementing them, tuning them, debugging them, and understanding how numerical choices affect empirical behavior. Our project repeatedly showed that the mathematical objective is only one part of the final model. The realized portfolio also depends on the data construction, the rolling estimation window, the regularization parameters, the turnover penalty, the solver settings, the constraints, and the evaluation design. In that sense, the project became an applied study of numerical

optimization as a complete research pipeline rather than a purely theoretical exercise.

We are proud of the final shape of the project because it became an investigation of how optimization behaves when theory meets noisy and incomplete data. The equity results showed that the same mean-variance framework can look weak under conservative regularization but much stronger after coherent recalibration. The prediction-market results showed that even when an optimizer produces positive objective values and disciplined allocations, those outputs do not automatically translate into realized economic value. This was an important lesson: optimization can organize decision-making under uncertainty, but it does not remove the need for careful validation. The project also reflects the iterative development process expected for the final deliverable. We began with a broad idea, narrowed it into an implementable equity optimizer, tested different parameter regimes, extended the framework to daily data, and then adapted the logic to prediction markets. Along the way, we revised the code, generated diagnostics, rethought the interpretation of our results, and became more careful about what the available data could and could not support. The final report, code, figures, and self-critique therefore document not only the final model, but also the process by which the model evolved.

In the end, the project brought together several strands of work: a dynamic equity optimizer, rolling mean and covariance estimation, long-only constrained portfolio choice, turnover-penalized optimization, PyTorch-based gradient methods, monthly and daily backtests, benchmark comparisons, sensitivity analysis, and an extension of the same allocation logic to prediction markets. The equity portion showed how numerical calibration, regularization, rolling-window design, and solver behavior can determine whether an optimizer behaves like a near-passive benchmark or a genuinely active allocation rule. The prediction-market portion showed how the framework can be adapted to a very different setting, where prices represent market-implied probabilities and where execution, liquidity, event concentration, and chronology-safe validation become central. Taken together, the project became an intellectually fulfilling exercise in the main themes formulating optimization problems,

implementing them numerically, tuning them carefully, and evaluating the full empirical pipeline. We hope to explore some of these research directions ourselves, and we also hope the project provides a useful foundation for future researchers interested in applying numerical optimization to financial markets, prediction markets, and other noisy real-world decision problems.